

AI-Based Identity & Document Screening

Hackathon solution overview / Synthetic demonstration

96.15%

VIZ exact field accuracy
150 / 156 fields, synthetic

92.99%

Tamper CNN F1
100 fresh procedural images

4.01 s

Warm CPU median
6 calls, pipeline only

Problem aur proposed impact

Document fields, face evidence aur tamper signals ko alag-alag check karna review ko slow aur inconsistent bana sakta hai. Sentinel ek officer workspace mein evidence aur reasons dikhata hai. Intended benefit: more consistent review. Real checkpoint impact abhi measure nahi hua; officer hi final decision record karta hai.

Layer-wise implementation

L0: OpenCV crop, perspective aur quality gate. **L1:** Tesseract MRZ/VIZ, raw + corrected fields. **L2:** check digits, cross-field/date rules, fictional lookup. **L3:** classical heatmap aur separate synthetic U-Net. **L4:** YuNet/SFace comparison, inconclusive liveness evidence. **L5:** YAML-weighted points, missing evidence aur reasons. **L6:** React review, human decisions aur hash-chain audit. L2/L3/L4 parallel run hote hain.

Engineering contribution aur feasibility

Prototype raw OCR corrections ko traceable rakhta hai, heatmaps ko explainable points se connect karta hai, aur unavailable evidence ko explicitly dikhata hai. Synthetic CNN risk score se excluded hai. FastAPI, React/Vite/Tailwind, SQLite/SQLAlchemy aur CPU inference local laptop par tested hain. Separate functions aur REST routes module demos allow karte hain. Docker config ready hai; engine runtime unverified hai.

Evaluation se kya seekha

26 synthetic TD3 images par MRZ field accuracy 90.38%, valid-document all-check pass 22/25. Fresh CNN confusion: TN 16, FP 4, FN 7, TP 73. Copy-move IoU 0.179; classical recall 1.25% at fixed 0.55 index. Face ROC sirf two known generated identities ke 32 correlated pairs par illustrative hai. Process-cold 12.51 s; warm median above excludes HTTP, audit and UI. Browser performance varies with system load.

Privacy aur deployment limits

Uploaded images, OCR identity values aur face embeddings persist nahi hote. Opt-in audit scores, hashes, officer labels aur notes store karta hai. Owner chain rewrite kar sakta hai; external checkpoint chahiye. No production auth, verified liveness, genuine issuer stamp model or chip trust validation. Synthetic-only metrics real-world accuracy ya DPDP compliance establish nahi karte.

Next validation

Authorized diverse layouts/devices, independent face captures, validation-only threshold calibration, locked external testing aur retention/access controls production se pehle required hain.